

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: OE & NA**

2. **Subject Name: Maneuvering & Control** **L-T-P: 3-0-0**
Credits:3

3. **Pre-requisites:**

4. **Syllabus and reference books:**

Syllabus:

Introduction; The Control Loop and Basic Equations of Motion; Motion Stability and Linear Equations; Analysis of Course keeping and Controls-Fixed Stability; Stability and Control; Analysis of Turning Ability; Free-Running Model Tests and Hydraulic Models; Nonlinear Equations of Motion and Captive Model Tests; Theoretical Prediction of Hydrodynamic Coefficients and Systems Identification; Acceleration, Stopping, and Backing; Automatic Control Systems; Effects of the Environment; Vessel Waterway Interactions; Hydrodynamics of Control Surfaces; Maneuvering Trials and Performance Requirements; Application to Design; Design of Rudder and Other Control Devices

Reference Books:

- (1) THE DYNAMICS OF MARINE CRAFT Maneuvering and Seakeeping, Edward M. Lewandowski, World Scientific, Advanced Series on Ocean Engineering - Volume 22
(2) Guidance and Control of Ocean Vehicles, Thor I. Fossen, John Wiley and Sons

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	The Control Loop and Basic Equations of Motion	6
2.	Motion Stability and Linear Equations	4
3.	Analysis of Course keeping and Controls	4
4.	Analysis of Turning Ability	3
5.	Free-Running Model Tests and Hydraulic Models	4
6.	Nonlinear Equations of Motion and Captive Model Tests	6
7.	Systems Identification	3
8.	Hydrodynamics of Control Surfaces	6
Total number of hours		36

